Extracting DNA from Fruit

Biology 2e chapter 14 reading connection | Student worksheet / preparation

Name: _____ Class: _____ Date: _____

Before predicting

Extraction separates components of a sample. Detergent disrupts membranes; filtration removes large debris, and alcohol can precipitate nucleic acids. Visible material is a crude mixture, not a purity test or a direct measurement of DNA quantity.

Investigation question

How does the extraction mixture affect visible material recovered from fruit?

Prediction and reasoning:

Controls, independent variable, measurement and sampling plan:

How to use the virtual lab

Use the same preparation and observation time. Score visible strands: 0 none, 1 faint, 2 clear, 3 abundant. The virtual scale is ordinal; it is not DNA concentration.

Select the indicated tool in the equipment tray, then select its target on the apparatus. Follow the preparation checklist and observe the changes. Mash fruit with three tool passes; the no-fruit blank skips mashing. Add the assigned mixture, filter, and layer alcohol. Review observations, then click Collect data and continue. Complete the assigned comparison before exploring.

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Label every sample virtual or classroom. Keep independent replicates separate; retain unexpected results. Units: visibility score (0-3)

Condition / sample	Repeat	Source	visibility score (0-3)
Complete extraction	1		
Complete extraction	2		
Complete extraction	3		
No detergent control	1		
No detergent control	2		
No detergent control	3		
No fruit blank	1		
No fruit blank	2		
No fruit blank	3		

Equipment and observations

Record calibration, initial values, specimen selection, ambiguous observations or method changes. Use additional paper for larger classroom counts.

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Copy this page for each trial. Treatment: _____ Repeat: ____ Source: _____

minutes	visibility score (0-3)	Notes
0		
1		
2		
3		
4		
5		

Graph your observations

Label axes and units; keep each treatment and source identifiable. For repeated measurements, connect readings only from the same sample. For ordinal scores, explain the categories instead of treating the scale as concentration.

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Compare the visibility patterns and explain what each control tests.

Why can the visible precipitate not establish DNA purity or exact yield?

How is extraction different from replication? What additional evidence would test a replication model?

Explore after the comparison

Propose one additional test. Change one condition or sample choice, collect exploration separately, and explain what the new evidence adds.

Important limits: Illustrative visibility, not molecular replication or quantitative DNA yield. Fruit chemistry and detergent formulation vary. This activity supports DNA isolation; replication needs a separate evidence investigation.